A Project Stage-1
Report On

MACHINE LEARNING BASED EPILESPY IMAGE DIAGNOSIS SOFTWARE

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Kommuri Pratap Reddy Institute of Technology (Autonomous)

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In partial fulfillment of the requirements for the award of the degree of

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In

Computer Science and Engineering

By

P. ALEKHYA (23RA1A0513)

M. HARSHITHA (23RA1A0549)

B. ASHOK (23RA1A0519)

T. HARIPRIYA (23RA1A0546)

Under the guidance of

Mrs. H.Archana

M.Tech.

Assistant Professor

Dept.of CSE



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC AUTONOMOUS)

(Affiliated to JNTUH, Ghanpur (V), Ghatkesar (M), Medchal (D)-500088)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC AUTONOMOUS)

(Ghanpur (V), Ghatkesar (M), Medchal (D)-500088)

(AFFILIATED TO JNTU, Hyderabad)



CERTIFICATE

This is to certify that the Project Stage-1 entitled **"MACHINE LEARNING BASED EPILEPSY IMAGE DIAGNOSIS SOFTWARE"** being submitted by **P. Alekhya (23RA1A0513), M. Harshitha (23RA1A0549), B. Ashok (23RA1A0519), T. Haripriya (23RA1A0546)** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to the Kommuri Pratap Reddy Institute of Technology (Autonomous)(Affiliated to JNTUH, Approved by AICTE & Accredited by NAAC & NBA) is a record of confined work carried out by them under my guidance and supervision..

Guide's signature
Mrs. H .ARCHANA
M.Tech
Asst. Professor
Dept. of CSE

Signature of the Head of the Dept.
Dr. K.VAMSHEE KRISHNA
M.Tech, Ph.D.
Asst. Professor
Dept. of CSE

Place: Ghanpur

Date:

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC AUTONOMOUS)

(Ghanpur (V), Ghatkesar (M), Medchal (D)-500088)

(AFFILIATED TO JNTU, Hyderabad)



DECLARATION

We, **P. ALEKHYA (23RA1A0513), M. HARSHITHA (23RA1A0549), B. ASHOK (23RA1A0519), T. HARIPRIYA (23RA1A0546)** hereby declare that the project stage-1 report titled **"MACHINE LEARNING BASED EPILEPSY IMAGE DIAGNOSIS SOFTWARE"** under the guidance of Mrs. H. ARCHANA, Assistant Professor, Department of Computer Science and Engineering, Kommuri Pratap Reddy Institute of Technology, Ghanpur, is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this thesis have not been submitted to any other University and Institute for the award of any Degree or Diploma.

P. ALEKHYA	**(23RA1A0513)**
M. HARSHITHA	**(23RA1A0549)**
B. ASHOK	**(23RA1A0519)**
T. HARIPRIYA	**(23RA1A0546)**

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P. ALEKHYA **(23RA1A0513)**
M. HARSHITHA **(23RA1A0549)**
B. ASHOK **(23RA1A0519)**
T. HARIPRIYA **(23RA1A0546)**

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PSO3: Implement industry-standard principles, practices and strategies for software development lifecycle and possess proficient skills and knowledge of software design process including system modelling and performance optimization.

ABSTRACT

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. Early detection of epileptic seizures is important for timely medical intervention and improved patient care. Traditional seizure detection methods often require continuous monitoring and expert interpretation of Electroencephalogram (EEG) signals. To address these challenges, this project proposes a Machine Learning-Based Epilepsy Detection System Using EEG Data.

The proposed system utilizes EEG signals to identify patterns associated with epileptic seizures using two machine learning algorithms: Support Vector Machine (SVM) and Random Forest. The process involves collecting EEG data, preprocessing the signals to reduce noise, extracting relevant features, and training the machine learning models to classify the signals into normal and seizure-related categories. The performance of both algorithms can be evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to determine their effectiveness in seizure detection.

The system aims to provide fast and reliable predictions and can be extended to support real-time EEG signal analysis. By comparing the performance of SVM and Random Forest, the project seeks to identify a suitable model for automated seizure detection. The proposed approach has the potential to assist healthcare professionals in EEG analysis and continuous seizure monitoring. However, the actual performance depends on the quality and diversity of the dataset, and real-time capabilities require validation using streaming EEG signals. The system is intended as a decision-support tool and does not replace professional medical diagnosis.

Key Words: *Machine Learning, Epilepsy Detection, EEG Signals, EEG Signal Processing, Medical Image Processing, Data Preprocessing, Feature Extraction, Support Vector Machine, Random Forest, Classification, Seizure Detection, Pattern Recognition, Signal Analysis, Model Training, Model Testing, Disease Prediction, Medical Diagnosis, Healthcare System, Accuracy Evaluation, Automated Detection.*

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1. INTRODUCTION

Epilepsy is a neurological disorder characterized by repeated seizures caused by abnormal electrical activity in the brain. Early and accurate detection of epileptic seizures is important for providing proper medical care and treatment. Electroencephalography (EEG) is widely used to monitor brain activity and identify abnormal patterns related to epileptic seizures. However, manually analyzing EEG signals requires more time and medical expertise.

Machine learning techniques can help improve the epilepsy detection process by automatically analyzing EEG data. This project uses Support Vector Machine (SVM) and Random Forest algorithms to identify patterns associated with epileptic seizures. The system processes EEG data, extracts important features, and classifies the signals into normal and epileptic categories.

Epilepsy is a brain disorder that causes repeated seizures due to unusual electrical activity in the brain. During a seizure, a person may experience changes in body movements, behavior, awareness, or sensations. EEG signals are used to record brain activity and help identify abnormal electrical patterns. Early detection helps doctors provide appropriate treatment and care.

Machine learning is a technology that enables computers to learn patterns from data and make predictions. In this project, Support Vector Machine (SVM) and Random Forest algorithms are used to analyze EEG signals. The system preprocesses the data, extracts important features, and classifies the signals as normal or epileptic. The performance of the models is evaluated using accuracy, precision, recall, and F1-score.

1.1 Background of the Project

Epilepsy is a neurological disorder that affects the normal functioning of the brain and causes repeated seizures. It occurs due to abnormal electrical activity in the brain. Early detection of epilepsy is important because it helps doctors identify seizure-related conditions and provide appropriate treatment. Electroencephalogram (EEG) is commonly used to record the electrical activity of the brain and helps in identifying abnormal brain patterns.

In traditional methods, EEG signals are examined by medical professionals to identify normal and abnormal brain activities. This process may require considerable time and experience, especially when analyzing large amounts of EEG data. Manual analysis can also make it difficult to identify certain abnormal patterns accurately and consistently.

Recent developments in Artificial Intelligence and Machine Learning have introduced new possibilities for automating medical data analysis. Machine Learning algorithms can analyze EEG data, identify important patterns, and classify the data based on learned features. These techniques can help improve the speed and efficiency of epilepsy-related analysis.

The proposed project, Machine Learning Based Epilepsy Image Diagnosis Software, focuses on developing a machine learning-based system for identifying epilepsy-related

abnormalities using EEG data. The system uses preprocessing techniques to reduce noise and improve data quality. Important features are extracted from the processed data and provided to machine learning algorithms.

The proposed system uses Support Vector Machine (SVM) and Random Forest algorithms to classify EEG data into normal and epileptic categories. The system also provides prediction results and performance evaluation using measures such as accuracy, precision, recall, F1-score, and confusion matrix.

This project aims to develop an efficient and user-friendly software system that supports automated EEG data analysis and epilepsy-related classification. It is intended as a decision-support tool and does not replace professional medical diagnosis.

1.2 Project Overview

Machine Learning Based Epilepsy Image Diagnosis Software is developed as a web-based application that uses machine learning techniques to identify epilepsy-related patterns from EEG data. The system aims to support the analysis of EEG signals and provide prediction results through an easy-to-use interface.

The frontend provides an interactive user interface developed using HTML5, CSS3, and JavaScript. It allows users to access the application, upload EEG images or data, submit input for analysis, and view the prediction results.

The backend is developed using Python and the Django framework. It manages user requests, handles uploaded EEG files, performs data processing, connects the frontend with the machine learning models, and displays the prediction results.

The machine learning processing layer uses two algorithms: Support Vector Machine (SVM) and Random Forest. These algorithms are used to analyze the extracted features from EEG data and classify the input into epilepsy-related or non-epilepsy-related categories.

The system consists of the following major components:

1. **Dataset Module:** Collects and manages EEG datasets containing normal and epileptic samples for model development.
2. **Data Preprocessing Module:** Removes unwanted noise, cleans the input data, and prepares it for further analysis.
3. **Feature Extraction Module:** Extracts important features from EEG data to identify patterns useful for classification.
4. **Machine Learning Module:** Uses SVM and Random Forest algorithms to train and classify EEG samples.
5. **Prediction Module:** Analyzes the uploaded EEG input using the trained model and generates the classification result.
6. **Result Display Module:** Displays the prediction result through the web application in an understandable format.

The proposed system aims to make EEG data analysis easier and faster by combining machine learning algorithms with a user-friendly web application. The generated

results are intended to support medical analysis and are not a replacement for professional medical diagnosis.

1.3 Problem Statement

Epilepsy is a neurological disorder that affects the normal functioning of the brain and can cause repeated seizures. Identifying epilepsy-related patterns through EEG (Electroencephalogram) data is an important part of medical analysis.

Traditional EEG analysis is mainly performed by medical professionals through manual examination of brain signal patterns. This process can be time-consuming, especially when large amounts of EEG data need to be analyzed. Manual analysis may also involve difficulties in identifying complex patterns and maintaining consistent results.

Additionally, analyzing EEG images manually requires experience and careful observation to identify abnormalities associated with epileptic activity.

Therefore, there is a need for a machine learning-based system that can analyze EEG images and assist in identifying epilepsy-related patterns. The proposed system addresses this need by using Support Vector Machine (SVM) and Random Forest algorithms to process EEG data and generate prediction results through a user-friendly web application.

1.4 Objectives of the Project

The primary objectives of the proposed project are:

1. To develop a machine learning-based system for identifying epilepsy-related patterns from EEG images or data.
2. To collect and preprocess EEG datasets containing normal and epileptic samples for model development.
3. To implement Support Vector Machine (SVM) and Random Forest algorithms for EEG data classification.
4. To develop a web-based application that allows users to upload EEG input and view prediction results.
5. To improve the speed and efficiency of EEG data analysis by reducing the need for repetitive manual examination.

1.5 Scope of the Project

The scope of this project focuses on developing a machine learning-based epilepsy diagnosis support system using EEG images or data. The system uses machine learning algorithms such as Support Vector Machine (SVM) and Random Forest to analyze EEG patterns and generate classification results.

The application includes dataset management, data preprocessing, feature extraction, model training, prediction, and result display. It also provides a user-friendly web interface for uploading EEG input and viewing the generated results.

2. LITERATURE SURVEY

2.1 Overview

A literature survey helps to understand the existing research, techniques and methods used in a particular field. Epilepsy is a neurological disorder that causes repeated seizures due to abnormal electrical activity in the brain. Electroencephalography (EEG) is commonly used to record the electrical activity of the brain and identify abnormal patterns related to epileptic seizures.

Traditional epilepsy diagnosis mainly depends on the manual analysis of EEG recordings by neurologists. This process requires time, experience and careful observation. Machine learning techniques provide an effective approach for automatically analysing EEG data and classifying normal and epileptic seizure patterns.

This project, titled "Machine Learning Based Epilepsy Image Diagnosis Software", proposes a machine-learning-based system using Support Vector Machine (SVM) and Random Forest (RF) algorithms. The system analyses EEG data, performs preprocessing and feature extraction, and predicts whether the input belongs to a normal or epileptic seizure class.

2.2 Benchmark EEG Datasets

Several publicly available EEG datasets are used in epilepsy detection research. The following datasets are relevant to this project.

2.2.1 Bonn University EEG Dataset

The Bonn University EEG dataset was introduced by Andrzejak et al. (2001). It contains five sets of EEG recordings, commonly identified as A, B, C, D and E. These recordings represent healthy and epileptic brain activity. It is widely used for developing and evaluating machine learning algorithms for epilepsy detection.

2.2.2 CHB-MIT Scalp EEG Dataset

The CHB-MIT dataset is a publicly available EEG database containing long-term scalp EEG recordings from paediatric patients with epilepsy. It includes seizure and non-seizure recordings with corresponding seizure annotations. Researchers use this dataset to develop and evaluate automated seizure detection systems.

2.2.3 EEG Signal Dataset

EEG signal datasets contain electrical brain activity recorded through electrodes placed on the scalp. These signals can be processed to identify abnormal patterns. The extracted information can be used to train machine learning models for classifying normal and epileptic seizure activities.

2.3 Machine Learning Algorithms Used in Epilepsy Detection

Different machine learning algorithms are used to analyse EEG signals and identify epileptic seizure patterns. The following algorithms are relevant to this project.

2.3.1 Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning algorithm used for classification. It identifies an optimal decision boundary that separates different classes. SVM can classify EEG features into normal and epileptic categories. Kernel functions such as Linear and RBF can be used to handle different classification patterns.

2.3.2 Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple decision trees to produce a classification result. It uses different subsets of training data and features to improve prediction performance. Random Forest can identify important EEG features and classify seizure and non-seizure patterns.

2.3.3 Feature Extraction

Feature extraction is an important stage in EEG-based epilepsy detection. It converts raw EEG signals into meaningful numerical information. Statistical, frequency-domain and time-domain features can be extracted to help machine learning algorithms identify abnormal brain activity.

2.4 Review of Related Work

Several researchers have proposed different machine learning and deep learning approaches for automatic epilepsy seizure detection.

Duan et al. [1] proposed an automatic method for epileptic seizure detection based on Deep Metric Learning in 2022. The researchers used EEG signals from the Bonn University and CHB-MIT datasets. They developed deep feature extraction methods to distinguish seizure and non-seizure activities. Their work demonstrated the potential of automatic seizure detection using EEG data.

Ali, Angelova and Karmakar [2] proposed an epileptic seizure detection system using the CHB-MIT dataset in 2024. They used Random Forest classification and extracted multiple features from EEG recordings. Their study focused on patient-independent seizure detection, class imbalance and real-world seizure event detection.

Zhang et al. [3] presented a review of epilepsy detection and prediction methods based on EEG signal processing and deep learning in 2024. Their research discussed EEG preprocessing, feature extraction, publicly available datasets and differentdeelearning algorithms. The study highlighted the importance of improving EEG signal analysis and seizure prediction methods.

Ghosh, Singh and Powar [4] proposed a seizure detection approach using the CHB-MIT dataset through multi-domain feature engineering and classical machine learning

in 2026. They extracted time-domain, frequency-domain, wavelet and spatial features from EEG signals. They evaluated K-Nearest Neighbours, Support Vector Machine and Random Forest classifiers. Their research demonstrated the effectiveness of classical machine learning algorithms for EEG-based seizure detection.

Andrzejak et al. [5] studied nonlinear characteristics of brain electrical activity in 2001. Their research contributed to the development of the Bonn University EEG dataset, which is widely used for evaluating epileptic seizure detection methods.

The above research studies demonstrate the importance of machine learning and deep learning techniques in detecting epileptic seizures using EEG signals and images. Existing approaches mainly focus on EEG preprocessing, feature extraction, signal classification, and deep learning-based image analysis. Some studies use SVM and other machine learning algorithms, while others apply CNN and LSTM architectures for automated seizure detection. However, there is an opportunity to develop a simple, integrated web-based application that combines EEG input processing, machine learning classification, and user-friendly result presentation.

The proposed project, Machine Learning Based Epilepsy Image Diagnosis Software, addresses this opportunity by developing a web-based system using Support Vector Machine (SVM) and Random Forest algorithms. The system includes EEG dataset collection, preprocessing, feature extraction, model training, prediction, and result generation. It uses Python and Django for backend development and HTML, CSS, and JavaScript for the frontend interface. The proposed application aims to support the identification of epilepsy-related patterns and provide understandable prediction results through a user-friendly interface. The system is intended as a supporting tool and not a replacement for professional medical diagnosis.

3. EXISTING SYSTEM

The existing system for epilepsy detection mainly depends on Electroencephalography (EEG) signals to monitor and record the electrical activity of the human brain. EEG is a widely used medical technique that helps doctors observe brain activity and identify abnormal electrical patterns associated with epileptic seizures. It plays an important role in examining patients who experience repeated seizures and other neurological problems.

In the traditional method, doctors and medical professionals manually examine EEG recordings to identify unusual brain signal patterns. They observe changes in the frequency, amplitude, and shape of the signals to detect possible epileptic activity. This process requires medical knowledge, experience, and careful examination of the recorded data.

Manual EEG signal analysis is a time-consuming process, especially when large amounts of data are collected over long periods. Doctors need to examine different signal patterns to identify abnormal activities. Sometimes, complex EEG signals and unwanted noise make it difficult to recognize seizure-related patterns accurately.

The existing methods also have limitations in terms of processing speed and consistency. EEG recordings may contain noise caused by eye movements, muscle movements, and other disturbances. These unwanted signals can affect the analysis process and make it difficult to identify important brain activity patterns. Manual interpretation may also vary depending on the experience of the medical professional.

Therefore, traditional epilepsy detection methods have limitations in handling large EEG datasets efficiently and automatically. To overcome these challenges, machine learning techniques such as Support Vector Machine (SVM) and Random Forest can be used to analyze EEG signals and classify normal and epileptic patterns. These techniques provide a basis for developing an automated epilepsy detection system that supports medical professionals in analyzing EEG data.

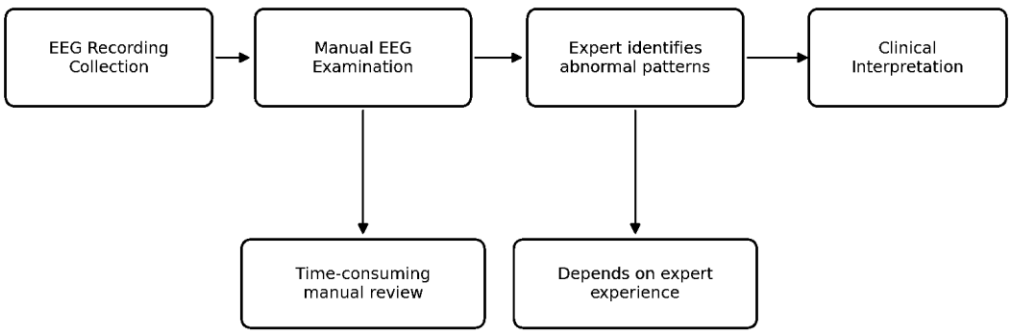


Fig. 3: Existing System

3.1 Limitations of the Existing System

The existing epilepsy detection approaches have several limitations that affect the accuracy and efficiency of EEG data analysis. Traditional epilepsy diagnosis mainly depends on medical professionals examining EEG signals and identifying abnormal brain activity. This process can be time-consuming, especially when a large number of EEG recordings or images need to be analyzed.

Manual examination of EEG signals requires specialized knowledge and experience. Identifying epilepsy-related patterns from complex EEG signals or images can be difficult, and the analysis may take considerable time. Different EEG formats and variations in signal quality can also create challenges during the examination process.

Another limitation is that some existing machine learning approaches focus mainly on individual stages such as EEG preprocessing, feature extraction, or classification rather than providing a complete integrated application. Some advanced deep learning methods also require large datasets and higher computational resources for model training and implementation.

Furthermore, many existing research approaches do not provide a simple user-friendly interface for uploading EEG data and viewing prediction results. Users may need separate tools for data processing, model execution, and result analysis. Therefore, existing approaches may not provide a complete and easily accessible solution for automated epilepsy-related pattern detection.

The proposed project aims to address these limitations by developing a machine learning-based epilepsy diagnosis software using Support Vector Machine (SVM) and Random Forest algorithms, with EEG data processing, classification, and result display through a web-based application.

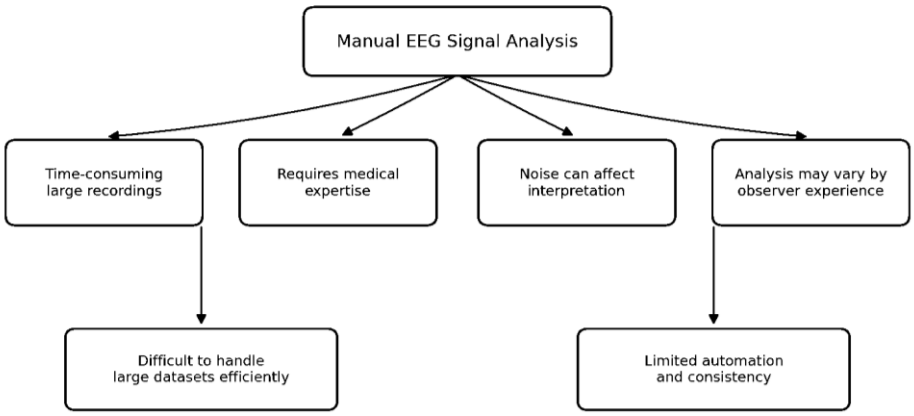


Fig. 3.1: Limitations of Existing System

3.2 Need for the Proposed System

The proposed system, Machine Learning Based Epilepsy Image Diagnosis Software, is designed to identify epilepsy-related patterns by analyzing EEG data or EEG images using machine learning techniques. The system aims to make the analysis process faster and easier compared to traditional manual examination.

The proposed system uses machine learning algorithms such as Support Vector Machine (SVM) and Random Forest to classify EEG patterns. Before classification, the uploaded EEG data or images are processed to improve their quality and prepare them for analysis. Relevant features are extracted and provided to the trained machine learning models for prediction.

The system provides a simple web-based interface where users can upload EEG data or images and receive prediction results. The frontend is developed using HTML, CSS, and JavaScript, while Python and Django are used for backend processing and connecting the machine learning models.

The system displays the classification result in an understandable format, helping users view the detected EEG patterns easily. It also reduces the time required for initial analysis and provides a consistent automated approach to pattern classification.

The proposed system aims to provide an efficient and user-friendly solution for epilepsy-related EEG pattern detection. It can support medical professionals in their analysis, but the prediction results must be reviewed by qualified healthcare professionals and should not be treated as a final medical diagnosis.

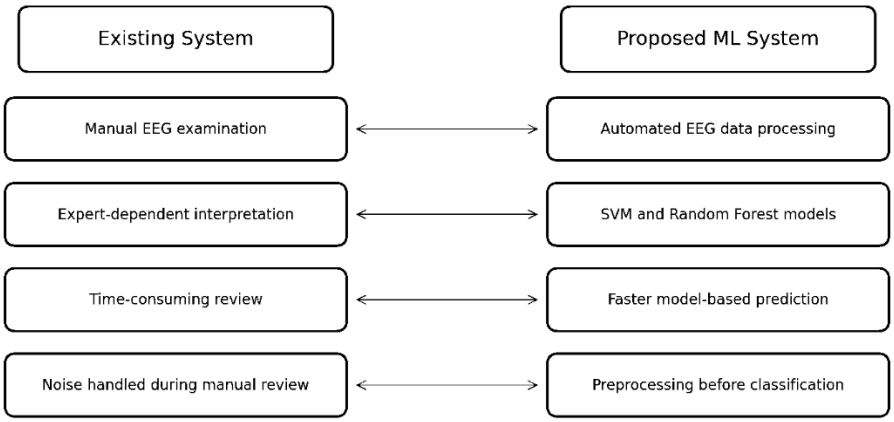


Fig. 3.2: Conceptual Comparision

4. PROPOSED SYSTEM

4.1 System Overview

The proposed system, titled Machine Learning Based Epilepsy Image Diagnosis Software, is a machine learning-based web application designed to identify epilepsy-related patterns by analyzing EEG data or EEG images. The system provides a platform for uploading EEG data, processing the input, performing machine learning classification, and displaying prediction results. It aims to reduce manual analysis time, improve processing efficiency, and support the identification of abnormal brain activity patterns.

The proposed system uses machine learning algorithms such as Support Vector Machine (SVM) and Random Forest. These algorithms are trained using an EEG dataset to learn the patterns associated with epilepsy. The system includes different stages such as dataset collection, data preprocessing, feature extraction, model training, classification, and result generation. Each stage plays an important role in preparing the input and producing prediction results.

The system processes the uploaded EEG data or images and extracts relevant features for classification. The trained SVM and Random Forest models analyze these features and predict whether the input contains epilepsy-related patterns. The prediction results are then sent to the application for display. This automated process helps users understand the analysis results in a simple and organized manner.

The frontend is developed using HTML, CSS, and JavaScript to provide an interactive and user-friendly interface. The backend is developed using Python and Django to manage application logic, input processing, database operations, and communication with the machine learning models. The trained models are integrated into the backend to generate predictions and display the results through the web application. The system is intended to support EEG analysis and does not replace professional medical diagnosis.

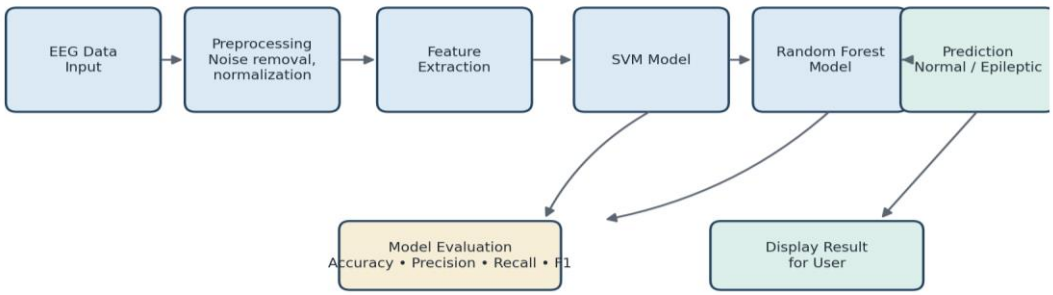


Fig. 4.1: Proposed System

4.2 Working Procedure

The proposed system follows an organized workflow to identify epilepsy-related patterns from EEG data or EEG images through a web-based application. Users access the application through the frontend interface and upload the required EEG data or images for analysis. The uploaded input is received by the backend for further processing.

The uploaded EEG data or images are processed through the preprocessing stage to prepare them for machine learning analysis. This stage helps improve the input quality and makes the data suitable for feature extraction. Relevant features are extracted from the processed input and passed to the trained machine learning models.

The machine learning models, Support Vector Machine (SVM) and Random Forest, analyze the extracted features and perform classification. Based on the learned patterns from the training dataset, the models generate a prediction indicating whether epilepsy-related patterns are detected in the input.

After classification, the prediction result is sent to the backend application. The Django backend manages the processing and communicates with the frontend interface. Finally, the result is displayed to the user through the web application in a simple and understandable format. The system is designed to support EEG analysis and not to provide a final medical diagnosis.

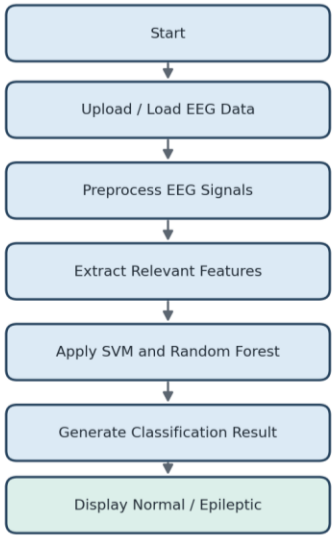


Fig. 4.2: Working flow

4.3 Advantages of the Proposed System

The proposed system provides several advantages by using machine learning algorithms to analyze EEG data or images. It helps reduce the time required for manual examination and provides an automated method for identifying epilepsy-related

patterns. The use of SVM and Random Forest algorithms supports the classification of EEG patterns based on the features learned during model training.

The system provides a simple and user-friendly web interface where users can upload EEG data or images and view prediction results. The integration of machine learning models with the Django backend makes the analysis process organized and easier to access. It also reduces the need to use separate tools for data processing, model execution, and result viewing.

The use of Python supports machine learning implementation and data processing, while Django manages backend operations and communication with the frontend. HTML, CSS, and JavaScript provide an interactive interface for users. The modular structure of the system makes it easier to maintain, test, and improve individual components.

Overall, the proposed system aims to save analysis time, reduce repetitive manual work, provide consistent automated classification, and make EEG pattern analysis more accessible. It can support healthcare professionals in reviewing EEG results, but the predictions must be evaluated by qualified medical professionals before any clinical decision is made.

5. SYSTEM ARCHITECTURE

5.1 Overall Architecture

The system architecture follows a structured machine learning-based approach to detect epilepsy using EEG (Electroencephalogram) signals. It integrates a systematic workflow that begins with EEG data acquisition, followed by data preprocessing, feature extraction, and classification. Machine learning algorithms such as Support Vector Machine (SVM) and Random Forest are used to analyze EEG signal patterns and classify them into normal and abnormal categories. The system includes a model evaluation process to measure classification performance using accuracy, precision, recall, and F1-score. A local storage mechanism is used to save processed data, extracted features, and trained models for efficient data management and faster execution. A graphical user interface (GUI) allows users to upload EEG datasets, train models, test data, and visualize prediction results. The real-time prediction module uses trained machine learning models to analyze new EEG samples and display the detection results. The overall architecture provides an organized and efficient approach to supporting epilepsy detection through machine learning techniques.

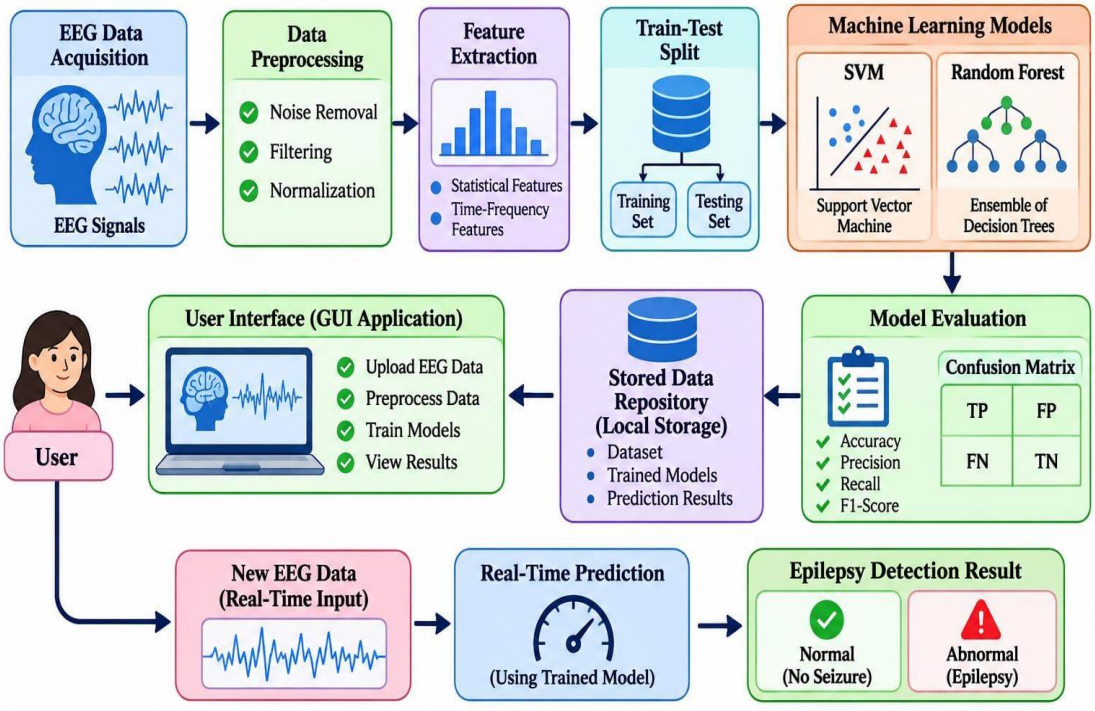


Fig.5.1: Overall system architecture

5.2 Frontend Architecture

The frontend architecture represents the user-facing part of the proposed application. It provides an interactive interface through which users can upload EEG data or images and view the epilepsy-related prediction results. The frontend is developed using HTML5, CSS3, and JavaScript.

HTML5 is used to define the structure of the web pages, while CSS3 is responsible for the visual appearance, layout, and responsive design. JavaScript manages user interactions, file uploads, form submissions, and communication with the backend.

The frontend allows users to access the application, upload EEG data or images, submit them for analysis, and view the results generated by the machine learning models. It also provides a suitable interface for displaying prediction information in a simple and understandable format.

The frontend communicates with the backend using HTTP requests through the Fetch API. This allows the uploaded input and prediction results to be exchanged between the user interface and the application services.

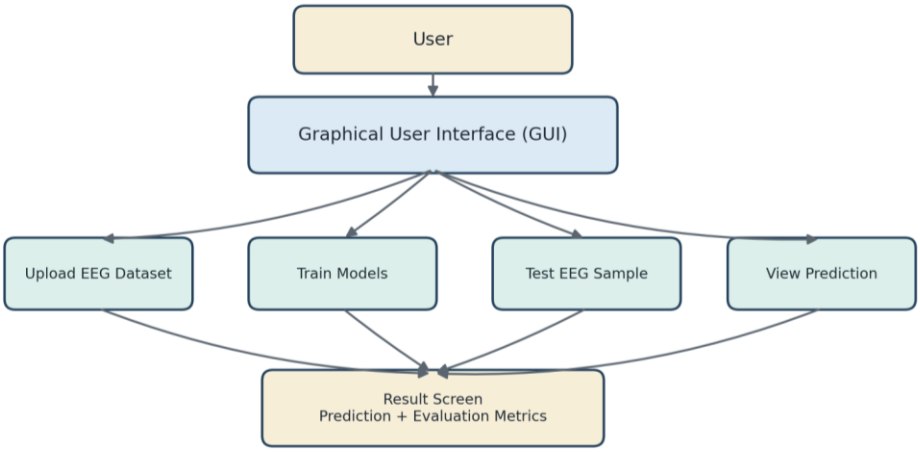


Fig. 5.2: Fronted architecture

5.3 Backend Architecture

The backend is developed using Python and Django and is responsible for managing the main application logic, processing user requests, handling uploaded EEG files, and coordinating communication with the machine learning models.

The Django backend acts as the communication layer between the frontend and machine learning components. It receives uploaded EEG data or images from the frontend and forwards them to the appropriate processing components. It also manages input validation, file handling, and prediction requests.

The preprocessing and feature extraction components prepare the uploaded EEG input for machine learning analysis. After preprocessing, the extracted features are passed to the trained SVM and Random Forest models for classification.

The backend receives the prediction results from the machine learning models and sends them back to the frontend for display. It also manages database operations, if a database is integrated, to store relevant application information and prediction records.

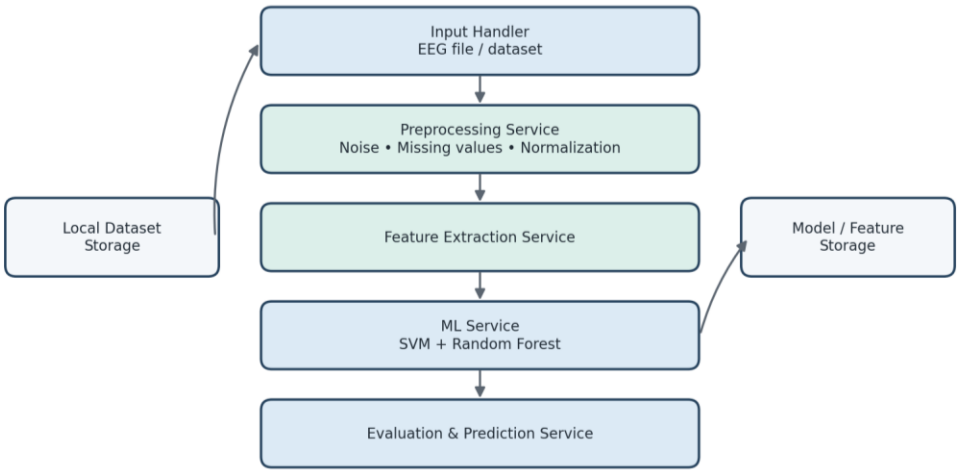


Fig. 5.3: Backend Architecture

5.4 Machine Learning Architecture

The Machine Learning Architecture is the main intelligent processing component of the proposed system. It uses machine learning algorithms to analyze EEG data or images and identify patterns associated with epilepsy.

The system uses Support Vector Machine (SVM) and Random Forest algorithms for classification. The EEG dataset is collected and processed before training the models. Preprocessing helps prepare the input data, while feature extraction identifies useful characteristics required for classification.

The SVM algorithm learns patterns from the training dataset and separates the data into different classes. The Random Forest algorithm uses multiple decision trees to perform classification and generate prediction results. Both trained models are integrated into the Django backend.

When a user uploads EEG data or an image, the system processes the input and provides the extracted features to the trained models. The models analyze the input and generate a prediction indicating whether epilepsy-related patterns are detected.

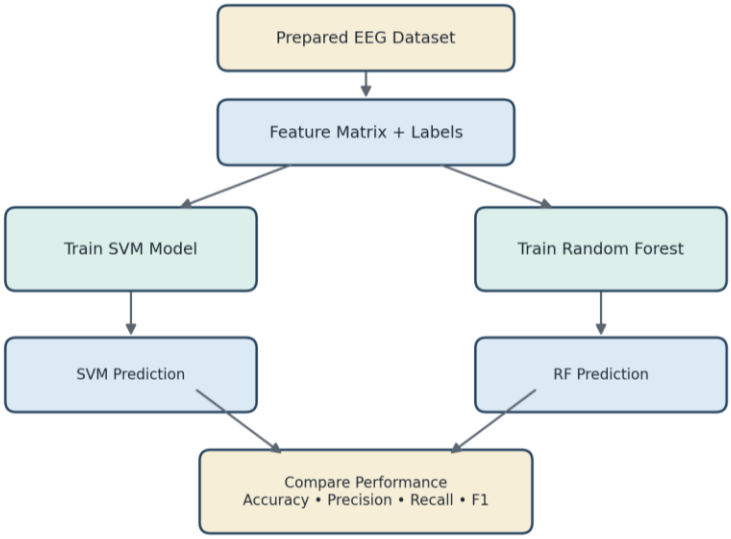


Fig. 5.4: ML Architecture

5.5 Database Architecture

A database can be used in the proposed application to store and manage relevant information required by the system. It may maintain user details, uploaded EEG file information, prediction records, and other application-related data.

The Django backend communicates with the database to perform operations such as inserting, retrieving, updating, and deleting records based on application requirements. This helps organize the information and maintain consistency.

The database is integrated with the Django backend, allowing application services to manage stored information without directly accessing the database from the frontend. Proper database organization also supports future improvements, such as maintaining prediction history and managing multiple user records.

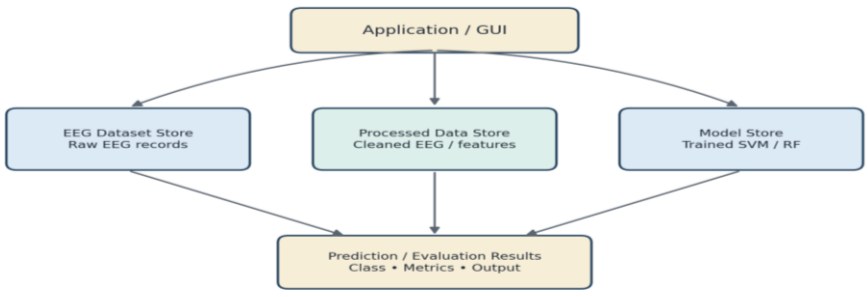


Fig. 5.5: Local Storage Architecture

6. MODULUS

The proposed system, Machine Learning Based Epilepsy Image Diagnosis Software, is divided into six major modules to organize the development and functionality of the application. Each module performs a specific task and works with the other modules to analyze EEG data or images and generate prediction results. The modules include the User Interface Module, Dataset Management Module, Preprocessing Module, Feature Extraction Module, Machine Learning Module, and Result Generation Module.

6.1 User Interface Module

The User Interface Module provides interaction between users and the epilepsy diagnosis application. It offers a simple and user-friendly interface through which users can upload EEG data or images and view the prediction results.

The frontend is developed using HTML5, CSS3, and JavaScript. HTML5 defines the structure of web pages, CSS3 manages the visual appearance and layout, and JavaScript handles user interactions and file uploads.

This module allows users to access the application, submit EEG inputs for analysis, and view the results generated by the machine learning models. It acts as the main access point for interacting with the system.

6.2 Dataset Management Module

The Dataset Management Module is responsible for collecting and managing the EEG dataset required for machine learning model development. The dataset contains EEG recordings or images used to train and test the classification models.

The dataset is organized into appropriate classes based on the available labels. Proper dataset management helps maintain the input information required for model training, testing, and evaluation.

This module provides the necessary data for developing the epilepsy-related pattern classification system.

6.3 Preprocessing Module

The Preprocessing Module prepares the collected EEG data or images before they are provided to the machine learning models. It helps improve input quality and makes the data suitable for further analysis.

Depending on the input format, preprocessing may include noise removal, normalization, resizing, and data cleaning. These operations help reduce unwanted variations in the input data.

This module ensures that the EEG input is prepared in a suitable format for feature extraction and classification.

6.4 Feature Extraction Module

The Feature Extraction Module identifies important characteristics from the preprocessed EEG data or images. These features help the machine learning algorithms understand the patterns present in the input.

The extracted features are selected and prepared for classification. This module reduces unnecessary information and provides useful input to the trained machine learning models.

Feature extraction plays an important role in improving the classification process.

6.5 Machine Learning Module

The Machine Learning Module is the main intelligent processing component of the proposed system. It uses two machine learning algorithms, Support Vector Machine (SVM) and Random Forest, to classify EEG patterns.

The models are trained using the prepared EEG dataset. SVM learns to distinguish different classes based on the extracted features, while Random Forest uses multiple decision trees to perform classification.

When new EEG input is uploaded, the trained models analyze its extracted features and generate a prediction. This module helps identify patterns associated with epilepsy based on the training data.

6.6 Result Generation Module

The Result Generation Module is responsible for displaying the prediction results produced by the machine learning models. After classification, the prediction is sent to the Django backend and displayed through the frontend interface.

The result is presented in a simple and understandable format, indicating the predicted class based on the model output. Users can view the result after submitting their EEG data or images for analysis.

This module completes the system workflow by providing the prediction output. The results are intended to support EEG analysis and must not be considered a final medical diagnosis without professional evaluation.

Formatting tip: This shortened version is designed to fit approximately three pages using Times New Roman, 12 pt, 1.5 line spacing, and normal margins. The exact page count depends on your document settings.

7. UML DIAGRAMS

UML (Unified Modeling Language) diagrams are used to represent the structure and working process of a software system. These diagrams help developers understand the system's design, functionality, and interaction between different components.

7.1. Use Case Diagram

The Use Case Diagram represents the interaction between the user and the Machine Learning Based Epilepsy Image Diagnosis Software. It illustrates the major functions available to users, including accessing the application, uploading EEG data or images, preprocessing the input, extracting features, performing machine learning classification, and viewing prediction results. The user interacts with the system through a web-based interface to obtain the analysis output.

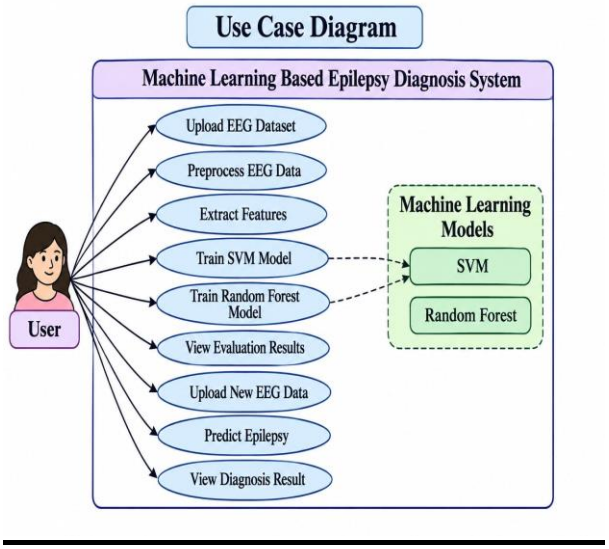


Fig. 7.1: Use Case Diagram

7.2. Activity Diagram

Activity diagrams are used to represent the step-by-step workflow of the proposed Machine Learning Based Epilepsy Image Diagnosis Software. They explain how the user interacts with the application, starting from uploading EEG data or images, followed by input validation, preprocessing, feature extraction, and machine learning classification using SVM and Random Forest.

The activity diagram also shows how the system generates the prediction result and displays it to the user through the web application. It helps understand the sequence of activities, decision-making steps, and overall working process of the proposed system.

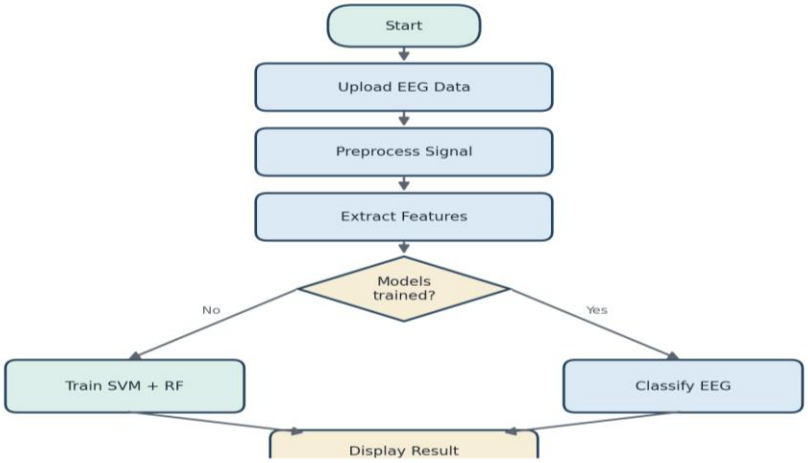


Fig. 7.2: Activity Diagram

7.3. Sequence Diagram

The Sequence Diagram illustrates the communication and sequence of interactions between the components of the proposed system. When a user uploads EEG data or an image through the frontend, the Django backend receives and validates the input. The backend forwards the input to the preprocessing component, where the required data preparation takes place. The feature extraction component identifies important characteristics and sends them to the trained SVM and Random Forest models. The models perform classification and return the prediction result to the backend. Finally, the result is sent to the frontend and displayed to the user.

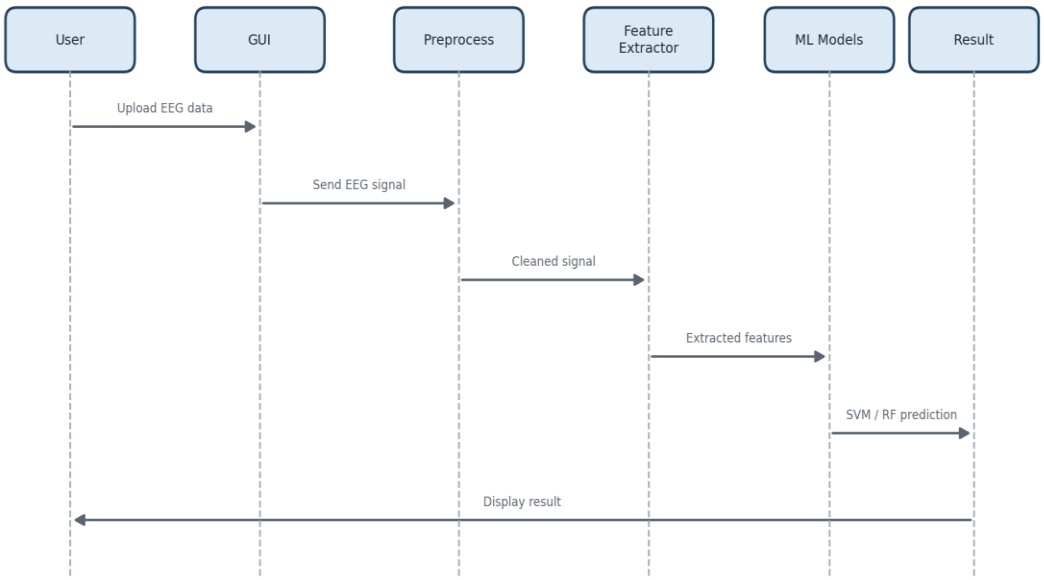


Fig. 7.3: Sequence Diagram

7.4. Class Diagram

The Class Diagram represents the structural design of the proposed Machine Learning Based Epilepsy Image Diagnosis Software. It illustrates the main classes, their attributes, methods, and relationships. The diagram includes classes such as User, EEG Input, Preprocessing, Feature Extraction, Machine Learning Model, SVM Classifier, Random Forest Classifier, and Prediction Result.

The User class manages user interactions, while the EEG Input class handles uploaded data or images. The Preprocessing and Feature Extraction classes prepare the input for classification. The SVM and Random Forest classes perform prediction using trained models. The Prediction Result class manages the classification output and provides the result for display.

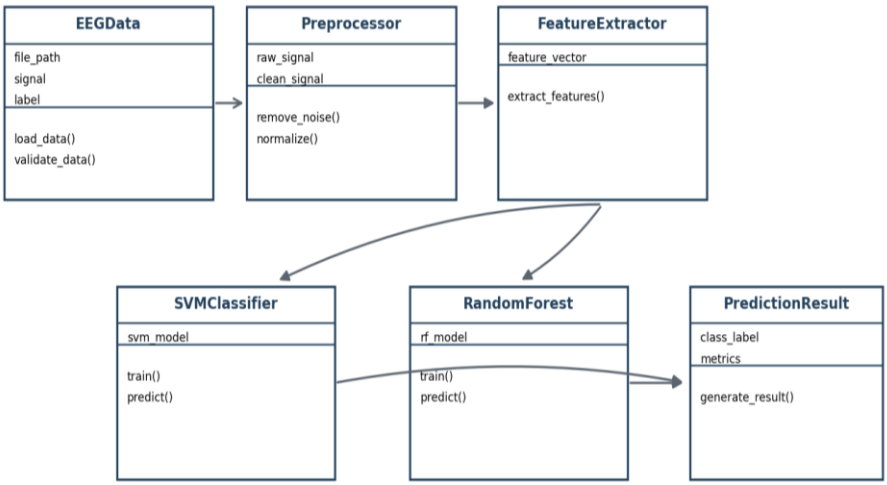


Fig. 7.4: Class Diagram

7.5 Component Diagram

The Component Diagram represents the major software components of the proposed epilepsy diagnosis system and their interactions. The Frontend Component provides the user interface using HTML, CSS, and JavaScript, while the Django Backend Component handles input requests, file uploads, application logic, and communication with the machine learning models.

The Machine Learning Component consists of the preprocessing, feature extraction, SVM, and Random Forest components. These components work together to analyze EEG data or images and generate prediction results. The Result Display Component receives the output from the backend and presents it to the user through the web application.

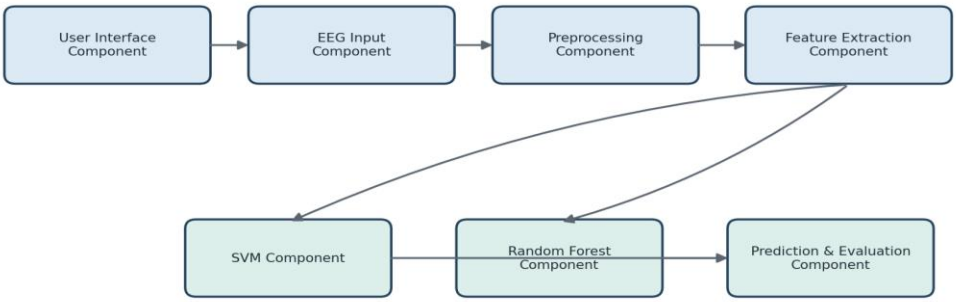


Fig. 7.5: Component Diagram

7.6 Deployment Diagram

The Deployment Diagram illustrates the physical and logical arrangement of the components required to run the proposed epilepsy diagnosis system. Users access the application through a web browser on a computer or other supported device. The browser communicates with the frontend interface, which sends requests to the Python Django backend.

The backend handles uploaded EEG data or images and coordinates preprocessing, feature extraction, and machine learning classification. The trained SVM and Random Forest models are integrated with the backend to generate predictions. The final results are returned to the web browser and displayed to the user. A database can also be connected to maintain user details and prediction records, if included in the implementation.

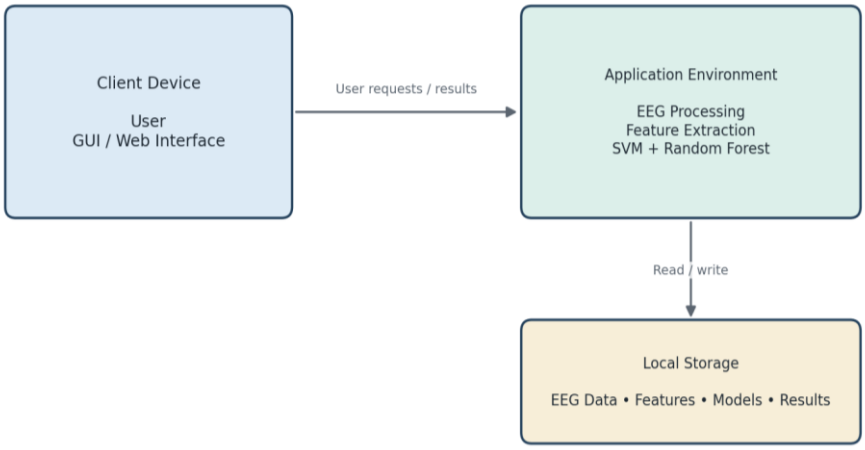


Fig. 7.6: Deployment Diagram

7.7 DATA FLOW DIAGRAMS

The Data Flow Diagram represents how information moves through the proposed Machine Learning Based Epilepsy Image Diagnosis Software. It explains how EEG data or images are uploaded, processed, analyzed using machine learning algorithms, and converted into prediction results**.**

7.7.1 Data Flow Diagram – Level 0

The Level 0 Data Flow Diagram represents the complete system as a single process. The user uploads EEG data or images to the system, and the application analyzes the input using trained machine learning models such as SVM and Random Forest.

The system generates the prediction result based on the analysis and displays it to the user through the web application. It provides an overall view of the interaction between the user and the system.

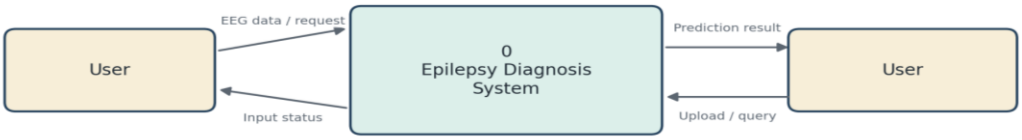


Fig. 7.7.1: Data Flow – Level 0

7.7.2 Data Flow Diagram – Level 1

The Level 1 Data Flow Diagram explains the major processes involved in the system. The uploaded EEG data or images pass through input validation and preprocessing. After preprocessing, the system extracts important features required for machine learning classification.

The extracted features are analyzed by the SVM and Random Forest algorithms. Finally, the system generates the classification result and displays it to the user through the frontend interface.

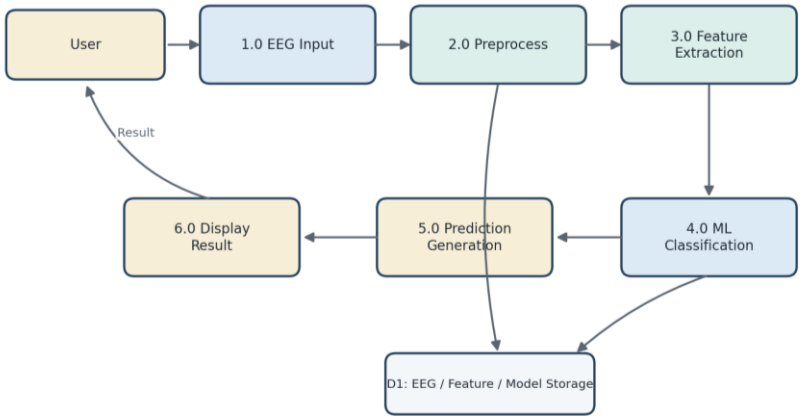


Fig. 7.7.2: Data Flow – Level 1

7.7.3 Data Flow Diagram – Level 2

The Level 2 Data Flow Diagram provides a more detailed explanation of the internal activities performed within the major processes of the proposed epilepsy diagnosis software. It explains how the system handles the uploaded EEG input from the initial stage to the final prediction output. The process starts when the user selects and uploads an EEG file or image through the frontend interface. The Django backend receives the input and performs file validation to ensure that the uploaded data is suitable for processing.

After validation, the preprocessing component performs the required operations according to the input format. For EEG signals, preprocessing may include noise removal and normalization, whereas EEG images may require resizing and image normalization. The prepared input is then passed to the feature extraction component, which identifies relevant characteristics needed for classification.

The extracted features are supplied to the trained Support Vector Machine and Random Forest models. The SVM model classifies the input by identifying patterns and separating different classes, while the Random Forest model uses multiple decision trees to generate its classification output. The machine learning component processes the extracted information and produces prediction results based on the trained dataset.

Finally, the prediction output is received by the Django backend and transferred to the frontend interface. The result is displayed to the user in a simple and understandable format. This detailed data flow explains the complete internal working process, starting from EEG input upload and ending with the generation and display of epilepsy-related prediction results.

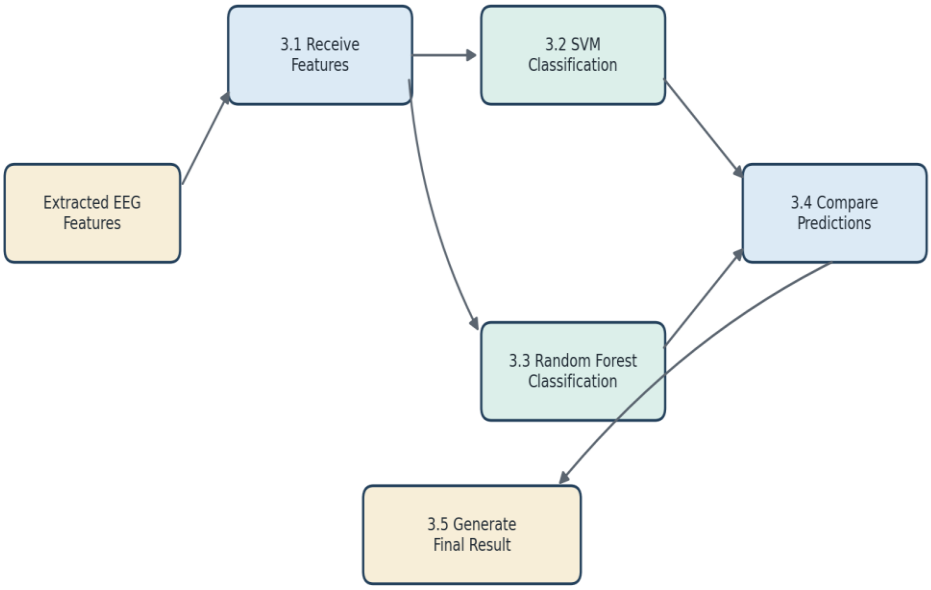


Fig. 7.7.3: Data Flow – Level 2

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